Jadeer
By Future Group

SALES ANALYSIS

Agenda

1. **Data Cleaning & Preparation**
   - Handling nulls and blanks
   - Standardizing date formats
   - Exporting cleaned dataset
2. **Data Warehouse (DWH) Modeling**
   - Star schema design (Dimensions & Fact)
   - Extracted DWH model
3. **Basket Analysis**
   - Data preparation for market basket analysis
   - Apriori algorithm application
   - Key metrics: Support, Confidence, Lift
   - Extracted basket rules
4. **Dashboard Development**
   - Sales performance dashboard
   - Market basket insights dashboard
   - Forecast analysis
5. **Validation**
   - Cross-checking Power BI values with Excel measures
6. **Final Takeaways & Business Impact**

📊 Data Cleaning and Data Warehouse Preparation Report

➢ Data Cleaning (Removing Nulls and Blanks)

Idea:
The raw dataset contained blank values in important columns such as `order_id`, `date format`, `city`, `branch`, and `product_category`.
We applied a **forward fill technique** where each null value is replaced with the last non-null value above it. This ensures continuity of order and product information. Additionally, we standardized the date format to maintain consistency.

Code:

```
import pandas as pd

# Load raw data
df = pd.read_excel("orders.xlsx")

# Replace blanks ("") with NaN to detect all the null and blancks
df = df.replace(r'^\s*$', pd.NA, regex=True)

# Columns to forward fill
cols_to_fill = ["order_id", "date format", "city", "branch",
"product_category"]
df[cols_to_fill] = df[cols_to_fill].ffill()

# Standardize date format
df["date format"] = pd.to_datetime(df["date format"], errors="coerce")
df["date format"] = df["date format"].dt.strftime("%Y-%m-%d %H:%M:%S")

# Save cleaned dataset
df.to_excel("orders_cleaned.xlsx", index=False)

print("Cleaning done!")
```

Benefit:

- Ensures there are no missing references for critical fields.
- Dates are standardized for time-series analysis.
- Data is now ready for dimensional modeling.

Extracted File:
➡️ `orders_cleaned.xlsx`

➢ Building Dimensions and Fact Table

Idea:
We applied a **star schema modeling approach** to transform the cleaned dataset into a Data Warehouse structure. This included creating:

- **Date Dimension (Date_Dim):** Contains granular details of dates (year, month, day, weekday, etc.) with a surrogate key starting from `1000000`.
- **Branch Dimension (Branch_Dim):** Contains unique branches and cities with a surrogate key starting from `100`.
- **Product Dimension (Product_Dim):** Contains product category, product name, and calculated unit price with a surrogate key starting from `1000`.
- **Fact Table (Fact_Orders):** The transactional data with foreign keys referencing the dimension tables.

Code:

```
# Date Dimension
df["date format"] = pd.to_datetime(df["date format"], errors="coerce")

date_dim = pd.DataFrame({
    "Full_Date": df["date format"].dt.date,
    "Day": df["date format"].dt.day,
    "Month": df["date format"].dt.month,
    "Month_Name": df["date format"].dt.strftime("%B"),
    "Quarter": df["date format"].dt.quarter,
    "Year": df["date format"].dt.year,
    "Weekday": df["date format"].dt.strftime("%A")
}).drop_duplicates().reset_index(drop=True)

# Add surrogate key starting from 1,000,000 - 1M
date_dim.insert(0, "Date_ID", range(1000000, 1000000 + len(date_dim)))


# Branch Dimension
branch_dim = df[["branch", "city"]].drop_duplicates().reset_index(drop=True)
branch_dim.insert(0, "Branch_ID", range(100, 100 + len(branch_dim)))


# Product Dimension
product_dim = df[["product_category", "product name", "quantity",
"total"]].copy()
product_dim["Unit_Price"] = product_dim["total"] / product_dim["quantity"]
product_dim = product_dim[["product_category", "product name",
"Unit_Price"]].drop_duplicates().reset_index(drop=True)
product_dim.insert(0, "Product_ID", range(1000, 1000 + len(product_dim)))
```

```
# Fact Table
fact_orders = df.copy()

# Adding Date_ID by merging with Date_Dim
fact_orders["Full_Date"] = pd.to_datetime(fact_orders["date format"]).dt.date
fact_orders = fact_orders.merge(date_dim[["Date_ID", "Full_Date"]],
on="Full_Date", how="left")

# Adding Branch_ID by merging with Branch_Dim
fact_orders = fact_orders.merge(branch_dim, left_on=["branch", "city"],
right_on=["branch", "city"], how="left")

# Adding Product_ID by merging with Product_Dim
fact_orders = fact_orders.merge(product_dim, left_on=["product_category",
"product name"],
                                right_on=["product_category", "product
name"], how="left")

# export the new dataset - the DWH model
with pd.ExcelWriter("DWH_Model.xlsx") as writer:
    date_dim.to_excel(writer, sheet_name="Date_Dim", index=False)
    branch_dim.to_excel(writer, sheet_name="Branch_Dim", index=False)
    product_dim.to_excel(writer, sheet_name="Product_Dim", index=False)
    fact_orders.to_excel(writer, sheet_name="Fact_Orders", index=False)

print("DWH done")
```

Benefit:

- Provides a structured **star schema** ready for BI tools (Power BI, Tableau, etc.).
- Improves query performance by separating dimensions and facts.
- Enables advanced analysis like **time intelligence, product performance, and branch comparison**.

Extracted File:
➡️ `DWH_Model.xlsx` (contains 4 sheets: `Date_Dim`, `Branch_Dim`, `Product_Dim`, `Fact_Orders`)

🛒 Basket Analysis Steps using Python

➢ Data Preparation

Idea:
We started by preparing the order dataset for market basket analysis. Each order can have multiple products, so we needed to transform the data into a format suitable for association rule mining.

Code (sample):

```
import pandas as pd

# Load data
df = pd.read_excel("orders_cleaned.xlsx")

# Keep only relevant columns
df = df[['order_id', 'product_id']]

# Create a basket format: each order as a row, each product as a column
basket = (df.groupby(['order_id', 'product_id'])['product_id']
            .count().unstack().reset_index().fillna(0)
            .set_index('order_id'))

# Convert values to 0/1 (whether product exists in the order) 0 for 0,1 is
mean true and false
basket = basket.applymap(lambda x: 1 if x > 0 else 0)
```

Benefit:
Prepared the data into a **basket format** (matrix of Orders × Products), ready for association rule mining.

Extracted file:
`basket.xlsx`

➢ Applying Apriori Algorithm

Idea:
We applied the Apriori algorithm to find frequent product sets (combinations that occur together across many orders).

Code (sample):

```
from mlxtend.frequent_patterns import apriori, association_rules

# Apply Apriori to find frequent itemsets
frequent_itemsets = apriori(basket, min_support=0.02, use_colnames=True)

# Generate association rules (support, confidence, lift)
rules = association_rules(frequent_itemsets, metric="lift",
min_threshold=1.0)

# export
rules.to_excel("basket_rules.xlsx", index=False)
```

Benefit:
Extracted **association rules** with key metrics:

- **Support:** How frequent the product combination appears.
- **Confidence:** Probability of buying product Y when X is purchased.
- **Lift:** Strength of the relationship (compared to random chance).

Extracted file:
`basket_rules.xlsx`

➢ Analysis & Insights

Idea:
We analyzed the rules to find meaningful product associations (e.g., which products are often bought together, strong combos for promotions, and opportunities for cross-selling).

Benefit:

- Identified top product pairs/triples frequently bought together.
- Provided insights into **customer purchase behavior**.
- Created a foundation for **recommendation systems** and **marketing campaigns**.

📊 Sales Dashboard Report

🔹 Page 1: Sales Analysis

🔸 Header Cards and Their Goals

Card	Description	Goal/Purpose
💰 **Total Sales**	Sum of all sales transactions	Shows overall revenue performance during the selected period.
📦 **Total Orders**	Total number of customer orders	Measures order volume and customer activity.
📊 **Total Quantity Sold**	Total units/items sold	Helps in analyzing product movement and stock turnover.
🧾 **Average Order Value**	Total sales / Total orders	Key indicator for customer spend behavior.
🏢 **Total Branches**	Number of active sales branches	Gives context to distribution and sales coverage.

🔸 Sales Trend Over Time

- **Purpose**: Tracks how sales evolved day by day from April 1 to May 31, 2021.
- **Insight**: Shows fluctuations in daily sales, with notable peaks and troughs.
- **Goal**:
  - Identify high-performing days or campaigns.
  - Detect sales drops that may require intervention.

🔸 Sales by City & Branch

- **Bar Chart by Branch Name and City**.
- **Insight**:
  - **Top Branch**: Downtown Cairo (3.16M)
  - **Next**: Corniche Alexandria (2.89M), Tanta (2.04M)
  - Branches like Maadi and New Cairo are underperforming.
- **Goal**:
  - Evaluate branch-level performance.
  - Guide strategic decisions like promotions or branch support.

🔸 Total Sales by Product Category

- **Top Performer**: Meal (6.28M)
- Followed by:
  - Sandwich (1.25M)
  - Side (973K)
- **Least Performer**: Dessert (44.4K)
- **Goal**:
  - Understand which categories are driving revenue.
  - Optimize product strategy.

🔸 Quantity Sold by Product

- **Top in Quantity**: Side (54K), Sandwich (52K)
- Lower quantities for high-price products like combos.
- **Goal**:
  - Reveal fast-moving products.
  - Helps in procurement and inventory planning.

🔸 Total Unit Price by Product

- Shows unit prices:
  - Highest: Product 25 (121.5)
  - Others: 87, 81, etc.
- **Goal**:
  - Understand pricing tiers and margins.
  - Support pricing optimization strategies.

🔹 Filters and Their Purpose

Filter	Function	Purpose
City	Filter by sales location	Analyze location-specific performance.
Branch Name	Select individual branches	Drill down to specific branches.
Product Category	Filter by product type	Compare performance across categories.
Date Range	Select timeframe	Perform time-based analysis for trends and comparisons.

📄 Page 2: Table View

🔸 Purpose

The table provides a **granular breakdown** of:

- **Total Quantity & Total Sales** per **Product Category**
- Grouped by **City**: Alexandria, Cairo, Tanta
- With sub-category/product breakdown (e.g., different combo IDs)

🔸 Insights & Goals

Feature	Value
🔍 **Detailed View**	Enables deep-dive into how specific products perform in each city.
🔁 **Comparative Analysis**	Compare performance of products across regions.
📦 **Inventory Control**	Identify over/under-performing products per city for restocking or promotions.
📈 **Strategic Planning**	Understand regional preferences to optimize offerings and pricing.

Example Insight:

- **Cairo** dominates sales in every category, with over **4.2M** total sales.
- **Meal** is the most sold product in all three cities, especially Cairo (2.8M).
- **Combo** has a smaller quantity but higher unit value.

📄 Page 3: Basket Analysis Dashboard

The **Basket Analysis** dashboard provides insights into product associations and purchasing patterns using **market basket analysis metrics** such as **Support**, **Lift**, and **Confidence**. These are commonly used to discover which products are frequently purchased together, helping with cross-selling strategies, promotions, and store layout planning.

Key Metrics:

- **Support**: Indicates how frequently an itemset appears in transactions.
- **Lift**: Measures the likelihood of a product being bought together with another compared to random chance.

- **Confidence**: Measures how often items in Y are purchased when items in X are purchased.

Visual Components:

- **Top 10 Products by Lift**:
  A bar chart that shows combinations of products (antecedent products) and how strongly they are associated with other products (consequent products). Higher lift values indicate stronger associations.
- **Top 10 Products by Support**:
  A horizontal bar chart showing which products frequently appear together in transactions. The higher the support, the more common the combination.
- **Support vs Confidence Scatter Plot**:
  This visual helps understand the relationship between product pair frequency (support) and the likelihood of joint purchases (confidence). Most combinations fall below 0.2 support and under 0.6 confidence, suggesting a wide variety of combinations but few dominant pairs.
- **Filter Options**:
  Sliders at the top allow dynamic filtering by support and lift ranges to focus on high-performing itemsets.

📄 Page 4: Forecast Analysis Dashboard

The **Forecast Analysis** dashboard provides a time-series overview of sales performance, including sales trends, order volume, quantity sold, and branch distribution over time.

Key Metrics:

- **Total Sales**:
  Cumulative revenue over the selected time period (April–May 2021).
- **Total Orders**:
  Total number of transactions/orders placed.
- **Total Quantity Sold**:
  Total number of units/items sold.
- **Average Order Value (AOV)**:
  Indicates the average revenue per order.
- **Total Branches**:
  Shows the business is operating across nine different locations.

Visual Components:

- **Sales Trend Over Time**:
  A line chart displaying daily or weekly fluctuations in total sales. Peak observed around mid-April and late May.

- **Total Quantity Sold Over Time**:
  Similar to the sales trend, this chart tracks the number of units sold over time. A visible dip in sales and quantity is seen at the start of June, indicating a possible drop-off or data cutoff.
- **Interactive Filters**:
  The dashboard includes filters by:
  - **City**
  - **Branch Name**
  - **Product Category**
  - **Date Range** (set from April 1 to May 31, 2021)

📄 Basket Analysis Tooltip:

This tooltip offers a **quick-glance summary of key basket analysis metrics** specifically for a particular product or itemset under the **Jadeer** brand. These metrics are likely shown when hovering over a specific data point in the main dashboard to provide instant insight into that combination's performance.

Metrics Explained:

- 🧮 **Support:**
  This value indicates that the product or combination appears in **all transactions** (or possibly XX transactions depending on the unit scale). This is **exceptionally high**, suggesting that the product or combo is **extremely popular** or even a core item in customer purchases.
- 📈 **Lift:**
  A lift value means that the likelihood of the consequent item being bought **increases XX times** when the antecedent item is purchased. This represents a **very strong association**, pointing to highly reliable product bundling behavior or dependent purchase patterns.
- 🔥 **Confidence:**
  This value suggests that when the antecedent product is bought, the consequent product is purchased how many time **over the time** (if expressed in percent) or possibly interpreted as a **scaled-up** confidence score. Either way, it represents **extreme certainty** in the rule – the presence of one item almost always guarantees the purchase of the other.

📄 Main Tooltip:

This tooltip provides a snapshot of high-level business performance indicators for the Jadeer brand.
It is designed to offer instant context when hovering over a specific product category, region, or

time period on the main dashboard—allowing for quick, informed analysis of sales dynamics and customer behavior.

Metrics Explained:

- 💰 **Total Sales:**
This represents the cumulative revenue generated, shown here as **$9.21 million**. It reflects overall commercial success and is a key top-line indicator of business health during the selected period.

- 📦 **Total Orders:**
A total of **39.20K orders** have been placed. This metric indicates transaction volume and helps assess customer activity and market demand.

- 📊 **Total Quantity Sold:**
This figure (**231.98K units**) shows the number of individual items sold, offering insight into sales volume irrespective of pricing—particularly useful for inventory planning and SKU-level performance reviews.

- 💳 **Average Order Value (AOV):**
With an AOV of **$91.64**, this metric shows the average dollar amount spent per order. It's crucial for understanding customer spending habits and for evaluating the impact of upselling or bundling strategies.

🧩 Tooltip for Table View Page: Enhanced Insight on Hover

🔸 Elements Shown in Tooltip

- **Sales by City & Branch (mini-bar chart)**
- **Sales Trend Over Time (mini-line chart)**
- **Unit Price (1.87K)**

🔸 Purpose and Power

Benefit	Description
🖱️ **Instant Insight**	Users get extra data without changing views.
📌 **Contextual Clarity**	Provides supporting charts that explain why a data point is high or low.
⏱️ **Time Efficiency**	No need to switch pages or apply filters — improves decision-making speed.

Benefit	Description
🔄 **Cross-Check Trends**	Quickly verify if a branch's performance spike is consistent with trends.

Example:
When hovering over the Downtown Cairo bar, the tooltip:

- Confirms it’s the top branch
- Shows consistent trend growth
- Indicates high unit price (adds value context)

📌 Final Takeaways

Component	Purpose
Cards	Give quick, high-level performance KPIs
Graphs	Provide trend analysis, product/category insights, and branch performance
Filters	Allow focused analysis by date, city, branch, or product type
Table View	Offers detailed, comparative, and product-level analysis per location
Tooltip	Adds hidden layers of insight with no disruption to workflow

---

🔍 Validation Report – Sales Analysis Dashboard

1- Total Sales

- **Card Value (Power BI):** 9,212,184.75
- **Excel Validation:**

```
=SUM(total)
```

- **Excel Expected Result:** 9,212,184.75 ☑

2- Total Orders

- **Card Value (Power BI):** 39,200
- **Excel Validation:**
- `=COUNTA(order_id)`
- **Excel Expected Result:** 39,200 ☑

3- Total Quantity Sold

- **Card Value (Power BI):** 231,965
- **Excel Validation:**
- `=SUM(quantity)`
- **Excel Expected Result:** 231,965 ☑

4- Average Order Value

- **Card Value (Power BI):** 91.64
- **Excel Validation:**
- `=SUM(total) / COUNTA(order_id)`
- **Excel Expected Result:** 91.64 ☑

5- Total Branches 🏬

- **Card Value (Power BI):** 9
- **Excel Validation:**
- `=COUNTA(UNIQUE(branch))`
- **Excel Expected Result:** 9 ☑

6- Sales Trend Over Time (⏰ Line Chart)

- **Power BI Visual:** Sales aggregated by day (`Full_Date`).
- **Excel Validation:** Insert **Pivot Table**:
  - Rows → `date format`
  - Values → `SUM(total)`
- Expected shape: daily trend matches Power BI. ☑

7- Sales by City & Branch (📍 Column Chart)

- **Power BI Visual:** Aggregated by `city + branch`.
- **Excel Validation:** Pivot Table
  - Rows → `city, branch`

- Values → `SUM(total)`
- Compare branch totals (e.g., **Cairo → Downtown 3.16M, Alexandria → Corniche 2.9M, Tanta → 2.0M**). ☑

8- Total Sales by Product (📦 Bar Chart)

- **Power BI Visual:** Aggregated by `product_category`.
- **Excel Validation:** Pivot Table
  - Rows → `product_category`
  - Values → `SUM(total)`
- Example:
  - Meal = 6.28M
  - Sandwich = 1.25M
  - Side = 973K
  - Drink = 458K ☑

9- Quantity Sold by Product (📊 Bar Chart)

- **Power BI Visual:** Aggregated by `product_name`.
- **Excel Validation:** Pivot Table
  - Rows → `product_name`
  - Values → `SUM(quantity)`
- Example:
  - Top side item ~ 54K
  - Next product ~ 52K ☑

10-Total Unit Price by Product (💲 Bar Chart)

- **Power BI Visual:** Unit price = `total ÷ quantity` per product.
- **Excel Validation:**
- `=SUM(total)/SUM(quantity)`

  in a Pivot Table grouped by `product_name`.

- Example:
  - Max product ~ 121.5
  - Min product ~ 42 ☑

🔮 Future Analysis & Next Steps

After completing the **Sales Analysis** and **Basket Analysis**, the next logical step is to expand the scope of analytics towards more predictive and prescriptive approaches. This will enable the business to move from *"what happened?"* to *"what will happen, and what should we do about it?"*.

➢ Customer-Level Analysis

- **RFM Analysis (Recency, Frequency, Monetary):** Identify high-value customers, loyal buyers, and churn risks.
- **Customer Segmentation:** Group customers into behavioral clusters (e.g., frequent snack buyers vs. full-meal buyers).
- **Churn Prediction:** Use historical purchase patterns to forecast customer attrition.

Benefit: Improves targeted marketing, retention campaigns, and personalized offers.

➢ Product-Level Analysis

- **Product Affinity & Recommendations:** Build a recommendation engine to suggest products based on basket patterns.
- **Price Sensitivity Analysis:** Study how price changes impact demand (elastic vs. inelastic products).
- **Seasonality Analysis:** Identify products with strong seasonal trends to optimize stock planning.

Benefit: Smarter inventory management and upselling opportunities.

➢ Time-Series & Forecasting

- **Sales Forecasting:** Predict sales volume for upcoming months using models like ARIMA or Prophet.
- **Demand Planning:** Estimate product demand per branch to optimize supply chain and reduce stockouts.

Benefit: Better financial planning and operational efficiency.

➢ Branch & City Analysis

- **Branch Performance Benchmarking:** Compare branches by sales, average basket size, and order frequency.
- **Expansion Insights:** Analyze geographic trends to identify new potential markets for branch expansion.

Benefit: Supports strategic growth decisions.

➢ Promotion & Campaign Effectiveness

- **Campaign Impact Analysis:** Measure sales uplift pre- and post-promotions.
- **A/B Testing:** Evaluate the success of different promotional strategies.

Benefit: Maximizes ROI on marketing and promotions.

➢ Advanced AI & ML Use Cases

- **Recommendation Engine:** Personalized product suggestions in real-time ordering systems.
- **Dynamic Pricing Models:** Adjust pricing based on demand, time, or location.
- **Chatbot Analytics:** Integrate purchasing insights into conversational AI channels.

Benefit: Enhances customer engagement and increases sales opportunities.